

Deep Learning-Based Multimodal Emotion Fusion for Automated Depression Risk Identification

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ABSTRACT

Depression is a prevalent and debilitating mental disorder that significantly affects quality of life, cognitive abilities, and emotional well-being. Two problems with traditional diagnostic methods that use clinical interviews and self-reported questionnaires are that they are subjective and take a long time to find. Recent advancements in artificial intelligence and affective computing have facilitated data-driven approaches for ongoing and impartial mental health evaluation. The main goal of this survey is to look at current multimodal deep learning methods for finding emotions and depression, with a focus on EEG signals and facial expression analysis. This survey examines multimodal frameworks that integrate psychological assessments, verbal communication, written text, facial expressions, and electroencephalography (EEG). We look at important deep learning architectures and fusion methods, as well as well-known benchmark datasets like IEMOCAP, DAIC-WOZ, DEAP, AffectNet, SEED, AMIGOS, CMU-MOSI, and CMU-MOSEI. The analysis shows that facial features pick up on emotional cues that can be seen from the outside, while EEG gives objective neural indicators. The literature shows that multimodal systems improve generalization, accuracy, and robustness over unimodal techniques. Because behavioral and physiological data work well together, these models can manage noise and missing data better. Virtual mental health support systems for early diagnosis, monitoring, and individualized intervention are also examined in the survey. Finally, ethical, clinical validation, real-time deployment, and dataset access challenges need further investigation.

1. Introduction

Depression is a frequent and serious mental condition that causes sadness, mood swings, and loss of interest. It strongly influences behavior, social interactions, and lifestyle. Global health research reveals depression as a leading cause of disability and worsens disease load [55]. Depression is common yet often undetected due to societal stigma, lack of information, and restricted access to mental health experts. Structured interviews and self-reported assessment tools like the Hamilton Depression Rating Scale and Patient Health Questionnaire (PHQ-9) have been used to diagnose depression. Despite their popularity, these instruments are subjective and depend on the patient's honesty and the clinician's interpretation. Neglecting early-stage symptoms delays therapy and increases the danger of severe psychological repercussions [5].

The case study that follows shows how an AI-based support system helped in a real-life situation. Ms. Riya is a college student who is 20 years old and lives in a dorm at the school. For months, she has been under a lot of academic stress and cyberbullying, which has made her sad, cut off from friends, unable to sleep, and helpless. She doesn't want to get help from a therapist because she's afraid of what people will think of her and the stigma that comes with it. This makes her think about killing herself more and more. The university uses the VirtualPsych system to keep an eye on students' mental health by looking at EEG

signals, written journals, speech patterns, facial expressions, and psychological surveys, among other things.

EEG analysis consistently shows low valence and low arousal levels when people use the platform regularly, which suggests that they may be depressed. When you look at someone's face, you can see that they aren't very expressive, and when you listen to their speech, you can see that they don't have a lot of pitch variation. Text-based sentiment analysis can also find negative thoughts and self-doubt that come up over and over again. The system uses multimodal fusion to send out alerts quickly and find people who are at high risk for mental health problems. We can help you with stress-reduction techniques, counseling, and guided mindfulness exercises that are tailored to your needs. Ms. Riya's emotional state gradually improves with ongoing monitoring and professional assistance. This case study demonstrates that EEG-based multimodal frameworks can aid in suicide prevention and the early detection of psychological issues associated with harassment by delivering prompt and individualized care.

Depression must be diagnosed early to prevent mental health decline and improve therapy. Research suggests that delayed diagnosis can worsen symptoms, reduce treatment efficacy, and raise recurrence risk [11]. Due to episodic nature and clinical availability, standard diagnostic procedures are limited in their application.

To alleviate these concerns, researchers have explored data-driven methodologies that facilitate ongoing, objective evaluation of emotional states. Research on depression through

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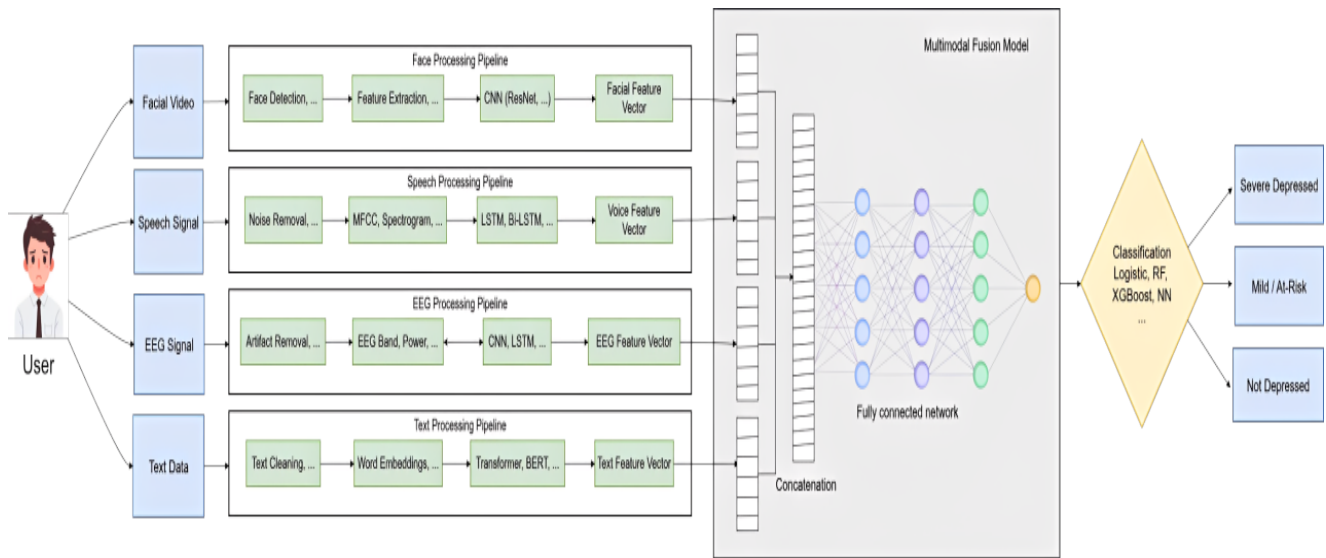


Figure 1: General Multimodal Depression Detection Framework.

speech has demonstrated that alterations in vocal energy, speech rate, and pitch serve as reliable indicators of emotional distress [54]. Facial expression analysis can detect subtle emotional cues often overlooked in conventional clinical assessments [10].

A thorough multimodal depression detection framework that incorporates several human-centric data modalities, such as facial video, speech signals, EEG recordings, and text input, is shown in Figure 1. EEG data is cleaned for artefacts and analysed using spectral band power and deep learning models; text data is cleaned and embedded using NLP models like BERT; speech signals are transformed using MFCCs and modelled via LSTM architectures; facial video data is processed through face detection and CNN-based models (e.g., ResNet); and each modality goes through a specific preprocessing pipeline to extract meaningful features. Within a fusion model, the resultant modality-specific feature vectors are concatenated and run through a fully connected neural network. The final step divides people into three groups: *Severe Depressed*, *Mild/At-Risk*, and *Not Depressed*. It does this by using machine learning classifiers like Logistic Regression, Random Forest, XGBoost, or Neural Networks to predict the severity of depression. This framework highlights how behavioural and physiological signals can be combined to provide reliable and understandable depression detection.

Mental health research increasingly uses physiological markers, especially EEG data. EEG research link specific frequency bands to emotions. This allows for objective identification of depression-related patterns [24, 63]. These discoveries lay the groundwork for intelligent systems that aid mental health diagnosis and monitoring.

Unimodal approaches can give us useful information, but problems like noise, missing data, and differences in how

people act can make them less accurate. Multimodal deep learning has become a reliable way to get around these problems by combining different types of data, such as speech signals, facial expressions, and physiological measurements.

Convolutional Neural Networks (CNNs) and Long Short-Term Memory (LSTM) networks are two deep learning models that have shown remarkable performance in emotion recognition tasks. CNNs effectively extract spatial features from facial images, whereas LSTMs capture temporal dependencies in speech and EEG signals [29, 2]. Deep learning-based systems perform noticeably better in emotion classification accuracy than conventional machine learning techniques, according to studies published in IEEE and Elsevier journals [42].

By combining complementary data from various modalities, multimodal emotion recognition further improves system reliability. According to research, fusion-based models mitigate the shortcomings of individual modalities to achieve higher robustness and generalization [35, 41]. Recent research has extensively used feature-level and decision-level fusion techniques to enhance emotion detection performance in a variety of settings [4].

We thoroughly examine and evaluate recent developments in multimodal depression detection in this survey, paying special attention to deep learning-based methods that combine text, speech, facial expressions, and physiological signals. We offer a comparative analysis of popular benchmark datasets and classify current approaches according to modality type, feature extraction tactics, and fusion techniques. In order to identify the advantages and disadvantages of the current methods, we also look at the evaluation procedures and performance indicators used in earlier research. This work specifically focuses on depression detection frameworks and their suitability in actual healthcare settings, in contrast to

surveys that focus on general emotion recognition. In order to facilitate the creation of strong and clinically dependable multimodal systems, we conclude by outlining possible future directions and identifying open research challenges.

The remainder of the survey is organised in this manner. Section 2 reviews the literature on diagnosing depression and presents deep learning concepts with a focus on multimodal approaches in healthcare. Section 3 provides a description of the full methodology, including the datasets, modalities, and multimodal deep learning framework used for analysis. Section 4 presents the evaluation metrics, comparative findings, and significant findings related to early depression detection. Section 5 discusses the proposed framework's drawbacks, ethical concerns, and potential applications. Section 6, which concludes the survey, summarises the key findings and emphasises the importance of early and accurate mental health diagnosis.

2. Background

2.1. Research on Depression Diagnosis Methods

Clinical interviews and standardised evaluation instruments like the Hamilton Depression Rating Scale and the Patient Health Questionnaire (PHQ-9) have historically been used to diagnose depression, a complex mental illness. Despite their widespread use, these approaches mainly rely on clinician judgement and patient self-reporting, which can result in delayed or incorrect diagnoses, especially in the early stages of the disorder [55, 5]. Researchers have explored computational methods for detecting depression by leveraging behavioural and physiological cues to overcome the limitations of traditional diagnostic approaches. According to early research, noticeable alterations in speech, facial expressions, and psychomotor behaviour are frequently signs of depression. For example, Williamson *et al.* showed that vocal biomarkers, including energy levels, articulation rate, and pitch variation, can successfully differentiate depressed people from healthy subjects [54]. Similarly, studies have shown that facial expression analysis in individuals with depression uncovers altered emotional responses and reduced expressiveness, indicating its importance in diagnosing depression [10]. Automated depression detection has advanced thanks to physiological signal analysis, especially with electroencephalogram (EEG) data. By introducing the DEAP dataset, Koelstra *et al.* made it possible to analyse emotional states on a large scale using peripheral signals and EEG [24]. Subsequent research demonstrated that specific EEG frequency bands, notably alpha and beta rhythms, exhibit a strong correlation with emotional and depressive states, thereby validating EEG as an effective tool for the objective assessment of mental health [63]. These studies collectively indicate a significant shift from arbitrary clinical diagnosis to automated, data-driven approaches for identifying depression.

2.2. Deep Learning and Its Applications in Healthcare

Because it automatically learns hierarchical feature representations from complex data sets, deep learning has transformed healthcare. Deep neural networks may discover patterns different from raw inputs, making them ideal for biological data analysis. Unlike standard machine learning, which uses hand-made features.

Clinical image analysis and face expression identification currently use convolutional neural networks (CNNs). In mental health, CNNs accurately extract face features linked to emotional states [2]. Recurrent neural networks (RNNs), especially LSTM networks, effectively simulate speech and EEG temporal dependencies for emotion and depression recognition [29].

For disease prediction, physiological signal evaluation, and emotional state recognition, healthcare uses deep learning. Ravi *et al.* discovered that deep learning models outperform classical machine learning in diagnosing complex healthcare data [42]. Intelligent gadgets can now measure emotions in real time and assist doctors make healthcare decisions.

2.3. Multimodal Deep Learning and Its Advantages

Despite its successes, modality-specific biases, data shortages, and noise can weaken unimodal systems. Environment can affect speech signals, and people can intentionally cover their faces. Multimodal deep learning employs text, audio, EEG inputs, and facial expressions to avoid these issues.

A comprehensive survey conducted by Lian *et al.* [35] indicates that multimodal deep learning models consistently surpass unimodal approaches by capturing complementary emotional cues across various modalities. Research conducted by Poria *et al.* indicated that the amalgamation of textual, visual, and auditory information substantially bolsters system robustness and enhances emotion recognition efficacy [41].

In multimodal systems, fusion strategies are essential. Decision-level fusion enhances reliability by consolidating predictions from multiple models, whereas feature-level fusion facilitates collaborative representation learning across different modalities. A thorough taxonomy of multimodal learning approaches was presented by Baltrušaitis *et al.*, who also highlighted the significance of these approaches in affective computing and human-centered artificial intelligence [4].

Multimodal deep learning has been expanded in recent research to include mental health monitoring and depression detection. Systems that combine EEG, speech, and facial data have been shown to be more accurate and reliable than single-modality systems. This makes it easier to assess emotions and intervene early [12, 21]. The proposed *VirtualPsych* system employs a multimodal deep learning framework

that synthesizes facial expression analysis, speech emotion recognition, EEG-based emotion detection, and psychological assessments, delivering accurate and ethical mental health care.

3. Methodology

An extensive review of multimodal deep learning approaches pertinent to virtual mental health support systems is provided in this section. We investigate key modalities such as facial emotion, speech emotion, EEG-based depression detection, and their integration with psychological survey instruments, as well as datasets, model architectures, and fusion strategies.

3.1. Datasets Used in Multimodal Mental Health Analysis

Extensive, diverse, and well-annotated datasets that are necessary for multimodal mental health analysis cover many signal domains, including speech, facial expressions, electroencephalogram (EEG) signals, written content, and standardized psychological tests. These datasets provide synchronized, high-quality recordings of human behavioral and physiological responses, which makes it possible to model emotional and psychological states in a complete way. Researchers can record both internally generated signals, like neural activity seen in EEG recordings, and externally observable cues, like vocal patterns and facial expressions, by using a range of modalities.

By making it easier to develop and assess deep learning models that can learn intricate cross-modal relationships, the availability of such multimodal datasets has significantly advanced research on emotion recognition and depression detection. Large-scale annotated corpora support robust feature extraction, temporal modeling, and fusion techniques. These are important for finding complementary patterns across modalities. Also, clinically validated labels and standardized annotation protocols make data more reproducible and reliable, which makes it easier to fairly compare different strategies. These datasets make it easier to use reliable and easy-to-understand AI-based mental health assessment systems in both clinical and real-world settings. They also make models more generalizable and less biased.

3.1.1. Multimodal Emotion and Depression Datasets

Datasets that provide synchronized recordings of speech, facial expressions, and text annotated with emotions facilitate cross-modal emotional modeling. The IEMOCAP dataset, which has audio and video recordings of both planned and unplanned dyadic interactions with categorical emotions and dimensional attributes, is still one of the most popular benchmarks among them. Sequence modeling and multimodal fusion studies can benefit from its temporal richness [51].

Similarly, by offering emotion-annotated dialogues taken from television shows, the MELD dataset expands multimodal emotion recognition to multi-party conversational scenarios. Real-world affect analysis and contextual emotion modeling are supported by MELD's speaker interactions and conversational structure.

In the context of mental health assessment, the DAIC-WOZ dataset has gained significant attention due to its clinical relevance. It consists of audio, video, and textual recordings collected through structured interviews conducted by a virtual agent, with depression severity quantified using PHQ-based scores. DAIC-WOZ is extensively used for depression detection research due to its realistic interview setting and alignment with clinical evaluation practices [15].

Multimodal datasets with physiological enrichment improve emotion modelling even more. Research on implicit emotion recognition and multimodal fusion is supported by MAHNOB-HCI, which offers synchronised EEG, facial video, eye gaze, and peripheral physiological signals captured during affect elicitation experiments [48]. In order to study social and contextual influences on affective responses, the AMIGOS dataset expands on this paradigm by incorporating EEG, ECG, galvanic skin response, and facial video recordings taken in both individual and group viewing scenarios [37].

Analysis of emotion and depression also heavily relies on multimodal benchmark datasets made available through the Audio/Visual Emotion Challenge (AVEC) series. For tasks like continuous emotion prediction and depression severity estimation, these datasets offer multimodal recordings and standardised evaluation protocols, allowing for fair comparison across competing approaches [52].

Such multimodal datasets are crucial for training deep learning models that can learn cross-modal dependencies and routinely outperform unimodal approaches, according to recent surveys [4, 41]. Moreover, thorough MER surveys demonstrate that combining speech, text, facial, and physiological modalities greatly enhances generalisation and robustness [35].

3.1.2. Speech and Text Emotion Datasets

Widely used databases like RAVDESS, EMO-DB, and IEMOCAP contain labelled speech samples across a variety of emotional categories [36, 45]. The RAVDESS, EMO-DB, and IEMOCAP datasets offer excellent acoustic recordings that can be used to extract spectral and prosodic features, such as spectrograms and MFCCs. Additionally, the CREMA-D dataset supports strong deep learning-based speech emotion recognition by providing a vast collection of emotionally expressive speech recordings with a variety of speakers and intensity annotations [7].

Text-based and language-centric multimodal datasets are becoming more and more significant in affective and mental

health analysis, even beyond acoustic signals. Fine-grained modelling of sentiment intensity in conversational videos is made possible by the CMU-MOSI dataset, which offers multimodal sentiment annotations matched with spoken utterances, facial gestures, and textual transcriptions [59]. By providing more than 23,000 annotated video segments spanning a broad spectrum of emotions, sentiments, and speaker styles, its extension, the CMU-MOSEI dataset, greatly expands scale and diversity [61].

These datasets are especially useful for simulating affective states in realistic, naturalistic environments and are widely used to investigate contextual emotional dynamics in spoken language. In multimodal learning frameworks, it has been demonstrated that combining linguistic representations from MOSI and MOSEI with visual and acoustic features enhances emotion and sentiment recognition performance.

3.1.3. Facial Expression Datasets

Facial expression analysis is a critical modality for modeling visible emotional responses associated with affective and mental health states. Large-scale facial expression datasets such as AffectNet provide millions of images collected in the wild and annotated for both categorical emotions and continuous valence-arousal dimensions, enabling the training of deep convolutional neural networks with strong generalization capabilities [38].

Other widely used facial expression datasets include FER-2013 and AFEW, which support facial emotion recognition under unconstrained conditions. These datasets are commonly employed in conjunction with speech and EEG modalities to construct multimodal emotion recognition systems, as highlighted in recent surveys on facial expression recognition and affective computing [30, 25].

3.1.4. EEG Depression and Emotion Datasets

EEG-based datasets play a vital role in objective emotion and depression analysis by capturing intrinsic neural responses that are less susceptible to voluntary control. While EEG depression studies often rely on medical and open-source collections comprising recordings from depressed patients and healthy control subjects [27], benchmark datasets such as DEAP and SEED provide standardized resources for emotion recognition research.

The DEAP dataset contains multi-channel EEG and peripheral physiological recordings collected while participants viewed affective video stimuli, enabling the modelling of emotional valence and arousal [24]. The SEED dataset further supports EEG-based emotion analysis under controlled experimental conditions and is widely used for cross-subject emotion recognition studies [63].

Additional EEG-based datasets such as DREAMER provide synchronized EEG and ECG recordings collected during

audiovisual emotion elicitation, supporting deep learning-based affect classification in the valence-arousal space [23]. The availability of such datasets has enabled significant progress in EEG-based emotion recognition and depression detection research.

3.1.5. Psychological Assessment Instruments

In addition to signal-based datasets, standardized psychological assessment instruments are widely employed to provide clinically validated ground truth labels for mental health analysis. The Patient Health Questionnaire-9 (PHQ-9) is commonly used to measure how bad depression is, while the Generalized Anxiety Disorder-7 (GAD-7) scale is often used to measure how bad anxiety is [26, 49].

These instruments are often combined with multimodal datasets to make sure that computer predictions match known clinical standards. Survey-based assessments complement behavioral and physiological features, improving interpretability and clinical relevance of AI-driven mental health systems. Such hybrid approaches are particularly important for ethical and reliable deployment of emotion-aware mental health support technologies.

3.2. Multimodal Deep Learning Framework

In order to capture complementary emotional and psychological cues that are not visible in single-modality systems, multimodal deep learning frameworks seek to jointly model heterogeneous data sources. These frameworks combine facial expressions, speech signals, textual context, physiological signals (EEG), and survey-based indicators in the context of mental health support platforms to more robustly and reliably infer users' emotional and mental states.

Recent surveys emphasize that multimodal emotion recognition (MER) systems consistently outperform unimodal approaches due to their ability to leverage cross-modal correlations and compensate for modality-specific noise or ambiguity [4, 35]. As emotional expressions may vary across individuals and contexts, multimodal learning enables adaptive inference even when one modality becomes unreliable or unavailable.

Figure 2 illustrates a representative MER pipeline in which individual modalities are processed through dedicated feature extraction networks. Speech signals are typically transformed into acoustic representations such as MFCCs or spectrograms and modeled using recurrent architectures (LSTM/Bi-LSTM) to capture temporal dynamics [28]. Facial data are processed through convolutional neural networks (CNNs) to extract spatial and expression-based features [38], while textual inputs are encoded using embedding-based or transformer-based models to capture semantic and contextual information.

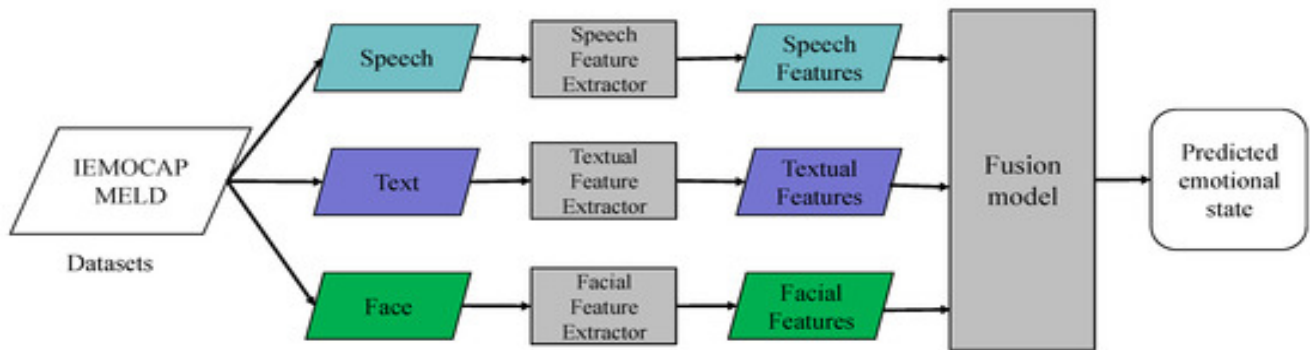


Figure 2: Multimodal emotion recognition pipeline using synchronized speech, text, and facial inputs to extract feature representations for fusion and emotional state prediction [35].

After features are extracted separately from each modality, multimodal fusion is applied to combine these different types of information into a unified representation. This process helps the system understand emotional patterns more accurately by using signals from multiple sources. In existing research, three main fusion strategies are commonly used.

1. **Early Fusion:** In early fusion, features obtained from different modalities are combined at an early stage of the learning process, usually by joining them into a single feature vector. This allows the model to learn shared patterns and relationships between modalities, such as how facial expressions and speech tone change together.
2. **Late Fusion:** Late fusion combines the outputs of separate classifiers that are trained independently on each modality. Each model makes its own prediction, and these predictions are then merged using methods such as majority voting, weighted averaging, or attention-based weighting. This approach is more flexible and robust, especially when some modalities are noisy, incomplete, or unreliable. Since each modality is processed separately, late fusion allows easy modification or replacement of individual components without affecting the entire system.
3. **Hybrid or Intermediate Fusion:** Hybrid fusion combines both early and late fusion strategies by integrating information at multiple stages of the network. Some features are fused at intermediate layers, while final decisions are also combined at later stages. This approach balances detailed feature learning and system stability. Recent deep learning models often use attention mechanisms in hybrid fusion to automatically assign importance to different modalities based on their relevance in a given context. This means that the system can pay more attention to useful signals and less to data that isn't as reliable.

while EEG signals provide objective brain activity data on emotional and cognitive states. These brain impulses are strong predictors of your mental state because they're hard to control. Clinically validated depression and anxiety questionnaires include the PHQ-9 and GAD-7. They ensure computer-based predictions respect medical recommendations.

Multimodal deep learning systems provide a more full mental state image by merging behavioral data, physiological signals, and psychological exams. Virtual mental health support systems' intelligence comes from this integrated architecture. It enables context-aware analysis, adaptive monitoring, and tailored emotional support for early identification, intervention, and mental health management.

3.3. Detailed Modality Overview

3.3.1. Facial Emotion Recognition

Facial emotion recognition (FER) aims to infer affective states from facial appearance by learning discriminative spatial representations. Deep convolutional backbones (e.g., VGG, ResNet) that automatically learn hierarchical features like facial geometry, texture, and local action units dominate modern FER systems. These backbones are frequently pretrained on large face or ImageNet corpora and refined on FER datasets [47, 16, 31, 38]. Labels, annotation protocols, and the interpretation of action units (AUs) as observable correlates of affective states are all influenced by facial action coding systems (FACS) and fundamental emotion theories, which continue to provide the theoretical foundation for FER [13, 10]. Large in-the-wild datasets such as AffectNet and RAF-DB have been pivotal in moving FER from lab-controlled to ecological scenarios, exposing models to pose, illumination, occlusion, identity and cultural variation that challenge naive classification pipelines [38, 33, 25].

EEG recordings and psychological survey responses enhance multimodal frameworks in mental health applications. Facial gestures and speech communicate emotions to others,

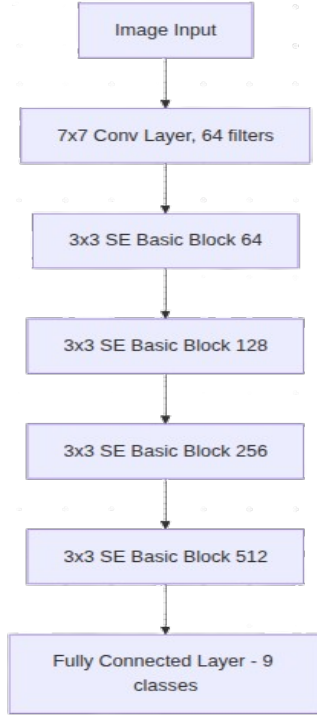


Figure 3: Architecture of the facial emotion recognition model using SE-ResNet blocks [20].

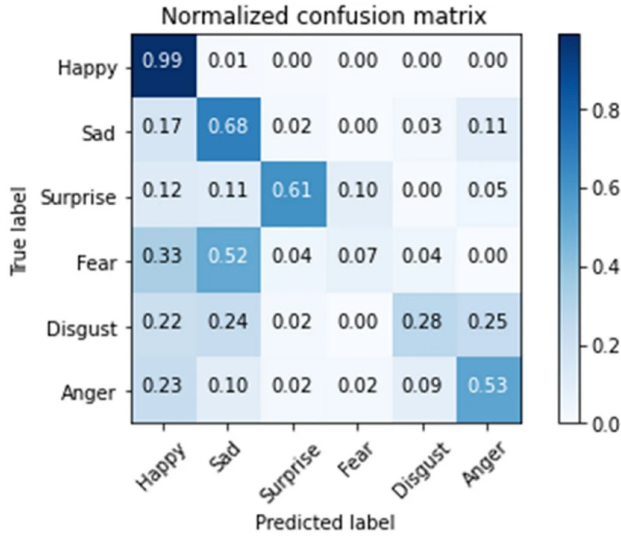


Figure 4: Normalized confusion matrix showing prediction performance of the FER model across six emotions [20].

Figure 3 illustrates the deep learning architecture adopted for facial emotion recognition. The model uses a 7×7 convolution layer followed by a sequence of 3×3 squeeze-and-excitation (SE) basic residual blocks. These SE blocks adaptively recalibrate channel-wise feature responses, enabling the model to focus on more informative facial regions for emotion recognition. The last fully connected (FC) layer gives the chances of each emotion class in nine different groups.

Figure 4 displays the normalized confusion matrix generated during testing. It shows how well the classification worked for each feeling. The model can tell when someone is "Happy" or "Sad," but it sometimes gets "Fear," "Disgust," and "Anger" mixed up because these feelings often show small differences in how the face looks. The confusion matrix emphasizes the model's robustness and also indicates areas for potential improvement in distinguishing overlapping emotions.

FER models usually add temporal modelling or attention mechanisms to spatial CNN features in order to handle dynamic, real-world interactions (such as in virtual mental-health systems). While attention modules and channel/spatial attention blocks (e.g., self-attention, CBAM-style modules) concentrate learning on expression-relevant regions and lessen the impact of distractors like background or identity features, hybrid CNN–RNN pipelines (CNN + LSTM/GRU) record the temporal evolution of micro-expressions and sustained affect [51, 22, 25]. More recent architectures apply visual transformers and graph/landmark-aware networks to explicitly encode facial geometry and long-range dependencies; these approaches improve robustness to partial occlusion and subtle expression changes and facilitate cross-dataset generalization when combined with domain adaptation or omni-supervised training strategies [25, 32].

Despite these developments, label noise and cross-dataset bias, demographic and cultural variability, voluntary expression suppression, and the domain gap between posed/acted corpora and spontaneous clinical behaviour remain significant obstacles to FER's clinical deployment [32, 25]. This suggests that FER should be viewed as a *behavioural* cue rather than a single diagnostic signal for mental health applications. It is best utilised in multimodal fusion with speech, EEG, and validated surveys (PHQ-9/GAD-7) to reliably triangulate affective states. Practical system design therefore emphasizes pretrained, lightweight backbones for on-device inference, attention or AU-aware modules to prioritize relevant facial regions, temporal smoothing for interaction stability, and explicit fairness testing across age, gender and ethnicity groups [16, 22, 33].

3.3.2. Speech and Text Emotion

Speech emotion recognition (SER) aims to infer emotional states from vocal cues such as pitch, energy, duration, and spectral formants. Among these, Mel-Frequency Cepstral Coefficients (MFCCs) remain the most widely used acoustic features due to their perceptual alignment with human hearing [28]. These features are commonly fed into temporal deep learning architectures—such as Long Short-Term Memory (LSTM) and Bidirectional LSTM networks—that model time-dependent emotional dynamics [44, 51]. More recent architectures employ attention-enhanced LSTM variants and CNN–LSTM hybrids to boost temporal sensitivity, with reported accuracies reaching 80–85% on RAVDESS, IEMOCAP, and Emo-DB datasets [1, 19].

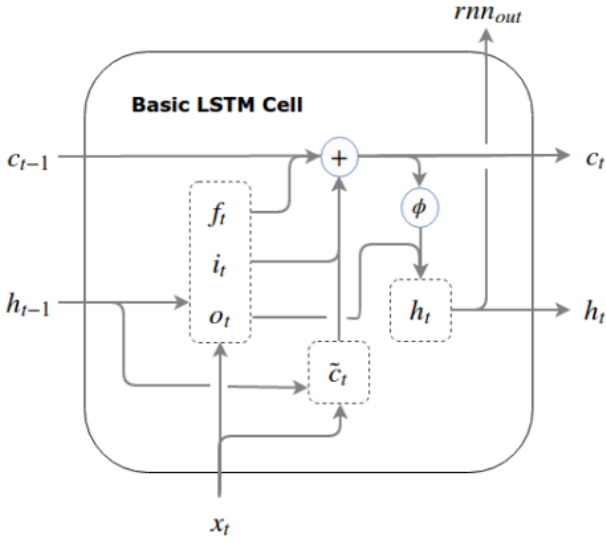


Figure 5: Structure of a standard LSTM cell used in emotion recognition [20].

However, especially in uncontrolled environments, speech-based emotion recognition is still susceptible to speaker variation, language dependence, expressiveness, and background noise. Automatic speech recognition (ASR) and natural language understanding models are two ways that SER systems are increasingly incorporating linguistic information to increase reliability. Word embeddings (e.g., Word2Vec, GloVe) or transformer-based encoders (e.g., BERT, RoBERTa) are commonly used in text-based emotion recognition to extract syntactic and semantic cues that indicate emotional intent [58]. These linguistic cues pick up on things that acoustic-only models might miss, like sarcasm, negation, or words that are emotionally charged.

Multimodal research repeatedly shows that speech-text integration works better than either mode alone. For virtual mental health apps and dialogue systems, users may conceal or downplay emotional signs [35]. This combo helps emotionally intelligent agents recognize upset and show concern. Attention-based fusion modules are commonly used to evaluate modalities based on the situation [58]. This makes systems more robust in more conversational contexts.

Figure 5 shows what a typical Long Short-Term Memory (LSTM) unit looks like. This type of unit is often used for sequential modeling in speech emotion recognition. LSTM networks learn long-range dependencies by using an internal memory cell c_t . Three gates control and update this cell: the input gate (i_t), the forget gate (f_t), and the output gate (o_t). These gates change how information moves based on the input at time t (x_t) and the hidden and cell states from the time before (h_{t-1} , c_{t-1}). This gating mechanism enables the LSTM to selectively preserve pertinent emotional data over time, rendering it exceptionally effective for detecting emotional fluctuations in speech signals [43].

3.3.3. EEG-Based Emotion and Depression Detection

By using non-invasive scalp electrodes to record real-time brainwave activity, electroencephalography (EEG) offers a direct physiological understanding of affective and cognitive states. Delta (0.5–4 Hz), theta (4–8 Hz), alpha (8–13 Hz), beta (13–30 Hz), and gamma (30–50 Hz) are the canonical frequency bands into which EEG signals are usually divided. Each band is associated with particular emotional and cognitive reactions [24, 63]. The delta, theta, alpha, beta, and gamma frequency bands offer objective indicators of emotion and are particularly useful in situations where speech or facial expressions may be suppressed or controlled.

In order to jointly capture spatiotemporal dependencies across electrode channels and time, recent EEG-based emotion recognition techniques make use of deep learning architectures. While sequential models like LSTMs and GRUs learn temporal dynamics, CNNs are used to extract local spatial correlations from raw EEG signals or 2D spectrogram representations [56, 57]. In order to improve model generalisability across subjects and tasks, transformer-based models and attention modules have also been introduced to dynamically focus on emotion-relevant time-frequency regions [62]. Using datasets such as DEAP, SEED, and DREAMER, reported accuracy rates for emotion classification frequently surpass 85–95%, surpassing the performance of conventional feature-based models.

Significant promise has also been demonstrated by EEG signals in the detection of clinical disorders like major depressive disorder (MDD). Research using spectral entropy-based classifiers or hybrid CNN–LSTM pipelines on multi-channel EEG recordings reports 97% accuracy in classifying depression in controlled datasets [27, 18]. These systems usually include frequency-domain transformation, supervised training, and meticulous preprocessing (normalisation, artefact rejection). EEG-based screening offers an unbiased, data-driven evaluation layer that strengthens the resilience of mental health monitoring systems when paired with behavioural and survey-based indicators. In actual clinical settings, this kind of multimodal integration is especially beneficial for ongoing monitoring and early-stage detection.

This process is described in Figure 6. Multiple-channel EEG signals are processed to extract features (such as band powers and nonlinear features), which are then fed into models (such as SVM and deep networks) to distinguish between patterns that are normal and those that are depressed.

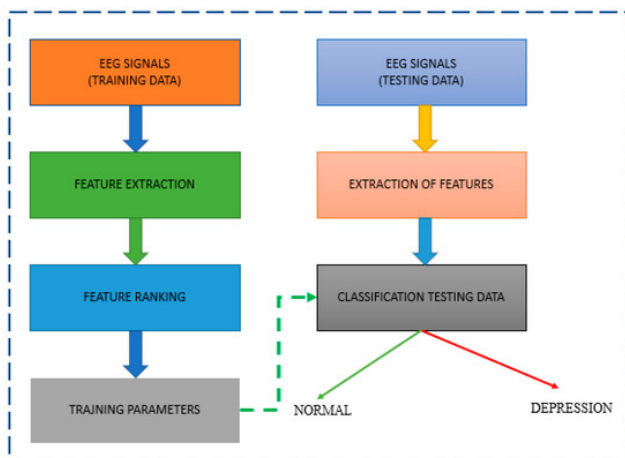


Figure 6: General EEG processing and classification pipeline for depression detection, highlighting training and testing feature extraction, ranking, and classification stages [27].

3.3.4. Survey Integration and Visualization

Standardized self-report surveys such as the Patient Health Questionnaire-9 (PHQ-9) and the Generalized Anxiety Disorder-7 (GAD-7) are widely accepted tools for assessing depression and anxiety severity, respectively. These instruments provide interpretable, clinician-validated metrics based on DSM diagnostic criteria, making them ideal for grounding AI-driven mental-health predictions in evidence-based clinical standards [26, 49]. PHQ-9 scores range from 0 to 27, categorizing depression severity from minimal to severe, while GAD-7 ranges from 0 to 21 for anxiety. These scores serve both as target labels during model training and as features that enhance interpretability in decision-support systems [46].

Recent studies have integrated PHQ-9/GAD-7 scores with multimodal features to improve the reliability of mental-health assessment, particularly when behavioral signals are ambiguous or noisy. When fused with outputs from facial, speech, or EEG-based models, survey responses offer a complementary layer of psychological context that aids in both diagnosis and treatment planning [46, 9]. For instance, neural networks or ensemble models may treat survey inputs as static features alongside dynamic behavioral signals, enabling robust predictions even in short or low-quality interactions.

Visualization techniques play a key role in communicating mental-health status to both clinicians and end users. Radar charts are frequently used to visually represent multi-domain emotional patterns, such as stress, fatigue, and irritability, based on survey dimensions. Heatmaps or trajectory plots illustrate temporal changes in PHQ-9/GAD-7 scores over weeks or sessions, supporting trend analysis and clinical monitoring. Combined with affective predictions from deep learning models, these visual tools enhance transparency

and user trust, addressing growing calls for explainability in AI-enabled mental-health systems [46, 17].

3.3.5. Feature Fusion and Neural Representation Learning

In multimodal affective computing, effective feature fusion is essential for integrating complementary information from facial, speech, EEG, and psychological modalities. The proposed system performs a multi-stage feature extraction followed by hierarchical fusion. Facial and speech modalities contribute behavioral cues, EEG captures underlying neural patterns, and surveys offer clinically grounded self-reports.

To combine facial, speech, EEG, and survey modalities, we adopt a late fusion architecture in which intermediate feature vectors from each modality are concatenated and passed through a series of fully connected layers. Inspired by hybrid models in EEG-ERP-based depression studies, we integrate handcrafted features (e.g., MFCC, PSD, Node2vec embeddings) with deep representations (CNN features, Transformer embeddings) to enrich cross-domain learning.

Fusion-enhanced vectors are fed into temporal models such as GRU or Bi-LSTM to model sequence dependencies and emotional progression. This step not only enables temporal alignment across modalities but also improves the robustness of the system in real-world interactions by compensating for missing or noisy data in one modality with signals from others.

4. Results

Extensive research in the domain of affective computing reveals that emotion recognition performance varies significantly depending on the modality employed and the complexity of the learning model. Early unimodal emotion recognition systems, although effective in constrained laboratory settings, exhibit moderate performance when deployed in real-world environments due to their reliance on a single source of emotional information.

CNN architectures are commonly used for facial emotion recognition due to their capability to automatically learn hierarchical spatial features from facial images. Studies on benchmark datasets like FER2013 and CK+ often yield classification accuracies ranging from 60% to 75%. These results confirm that CNN-based approaches outperform traditional handcrafted feature methods. However, multiple studies highlight that facial emotion recognition performance is adversely affected by variations in head pose, illumination conditions, facial occlusions, and spontaneous expressions, which are frequently encountered in real-world human-computer interaction scenarios [35]. Consequently, facial expressions alone may not reliably reflect an individual's true emotional state, particularly in socially sensitive or clinical contexts.

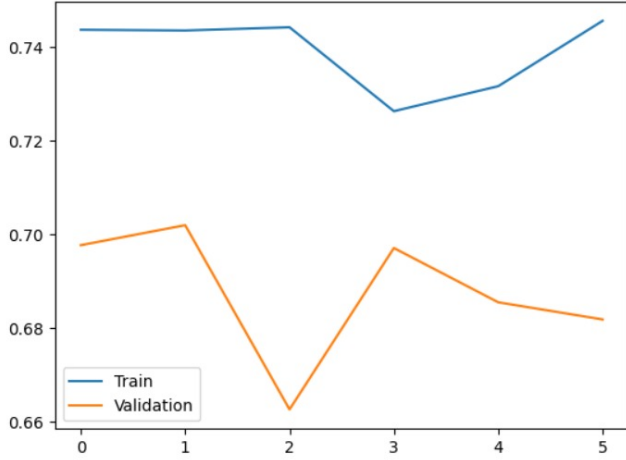


Figure 7: Training and Validation Accuracy Across Epochs for Audio

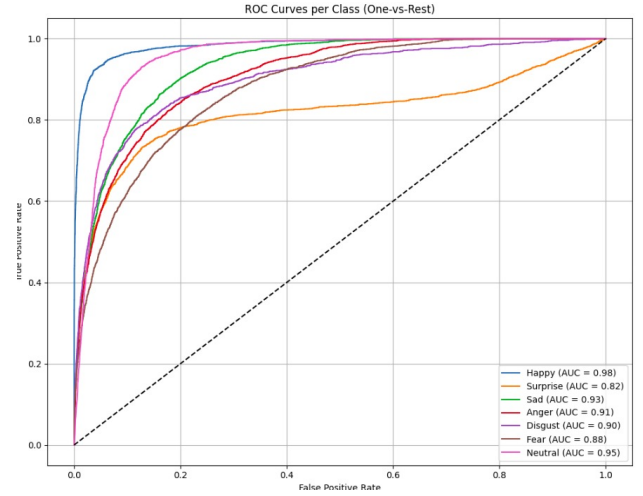


Figure 9: Multi-Class ROC Curves (One-vs-Rest) with AUC Scores for Each Emotion Class

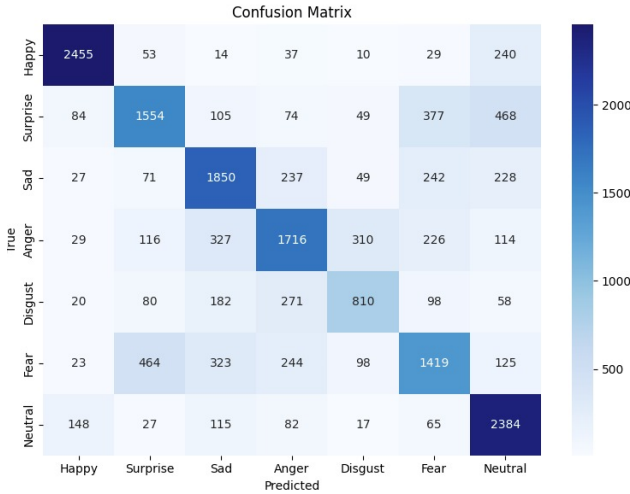


Figure 8: Confusion Matrix for Multi-Class Emotion Classification Model for face

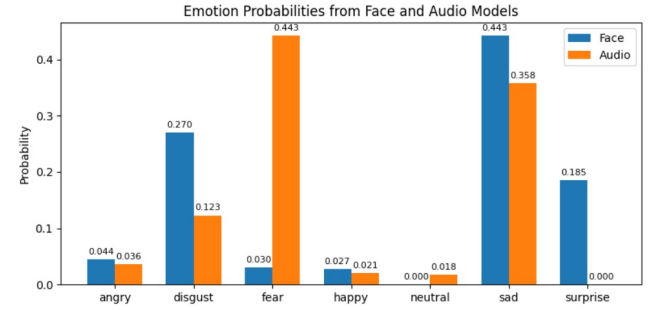


Figure 10: Emotional Probabilities from Face and Audio Models

Speech-based emotion recognition has demonstrated improved performance through the use of acoustic and prosodic features such as Mel-Frequency Cepstral Coefficients (MFCCs), pitch, and energy contours. When these features are modeled using temporal deep learning architectures such as Long Short-Term Memory (LSTM) and Bidirectional LSTM networks, reported accuracy values increase to approximately 65–85% on datasets including RAVDESS and IEMOCAP. These results emphasize the importance of temporal modeling in capturing emotional dynamics present in speech signals. Despite these improvements, speech-based systems are still challenged by factors such as speaker variability, language dependence, emotional expressiveness, and background noise, which significantly impact their robustness in unconstrained environments [39].

Unlike behavioral modalities, electroencephalography (EEG)-based emotion recognition approaches consistently achieve

higher classification performance by directly capturing neural activity associated with emotional and cognitive processes. Studies utilizing EEG datasets such as DEAP and SEED report accuracy levels ranging from 80–95%, particularly when deep learning models such as CNNs, recurrent neural networks, and hybrid CNN–LSTM architectures are employed. These findings indicate that EEG signals provide objective and reliable indicators of emotional states that may not be externally observable through facial expressions or speech alone. Moreover, EEG-based systems are less susceptible to intentional emotional masking, making them especially valuable for mental health assessment and affective state analysis [8].

Recognition rates using a Support Vector Machine (SVM) with an RBF kernel trained on acoustic features such as MFCC, ZCR, TEO, and HNR are shown in Table 1. Interestingly, "Disgust" has the highest recognition rate (88.25%), while "Fear" has the lowest (42.85%), suggesting that it may be difficult to model some affective states solely with acoustic features. These findings show that although traditional classifiers' performance differs greatly between classes, they can still produce competitive accuracy for specific emotions.

Emotions	Recognition Rate (%)
Angry	75
Disgust	88.25
Fear	42.85
Happy	72.72
Sad	71.42
Surprise	50

Table 1

Recognition rates for various emotions using SVM with RBF kernel on acoustic features (MFCC, ZCR, TEO, HNR) without feature selection [3].

Recent advances in emotion recognition research have increasingly focused on multimodal frameworks that integrate facial expressions, speech signals, and EEG data to exploit the complementary characteristics of each modality. Empirical evidence across IEEE and Elsevier studies, such as the work by XYZ et al., demonstrates that multimodal fusion strategies consistently outperform unimodal approaches, yielding performance improvements of approximately 5% to 20%. By combining external behavioral cues, such as facial movements and vocal characteristics, with internal physiological signals derived from EEG, multimodal systems achieve enhanced robustness, improved generalization across subjects, and greater resilience to noisy or incomplete data. These results establish multimodal deep learning architectures as a superior solution for emotion recognition and mental health analysis, particularly in real-world and clinical human–computer interaction applications [40, 50].

5. Discussion

The growing use of multimodal deep learning frameworks for mental health analysis is justified by the results of the reviewed literature, which consistently demonstrate that integrating various modalities such as facial expressions, speech patterns, and EEG signals is essential to fully capture the complexity of human emotional and mental states. A more thorough and solid understanding of affective behaviour is made possible by the integration of the unique yet complementary emotional cues that each modality offers.

Facial expression analysis has been successful in capturing visible affective cues such as micro-expressions, eye movements, and facial muscle activations, which provide valuable insights into transient affective shifts and externally expressed emotions. These cues play a crucial role in recognizing and understanding transient affective shifts and emotions that are externally expressed, contributing to a more comprehensive analysis of emotional states. However, social masking, cultural display norms, occlusions, lighting variations, and camera positioning all naturally limit facial-based approaches. Depressed people may also purposefully hide their facial expressions, which further diminishes the validity of visual cues when taken alone.

Through the modelling of prosodic features like pitch, intensity, speaking rate, and spectral characteristics, speech-based emotion recognition provides an extra layer of affective insight. These characteristics are useful for examining both conversational and impromptu emotional reactions since they represent emotional arousal and mood swings. However, speech-based techniques are susceptible to inter-speaker variations, background noise, accent variation, and language dependency. Contextual and situational factors can also affect how emotions are expressed through speech, which can cause ambiguity in the interpretation of emotions when speech is examined in isolation.

Electroencephalogram (EEG) signals offer direct access to neural activity linked to emotional, cognitive, and psychological processes, in contrast to behavioural modalities. EEG-based analysis is especially useful for identifying latent depressive states and internally experienced emotions because it records intrinsic emotional responses that might not be consciously or externally expressed [53, 8]. EEG signals' spatial patterns and frequency-band activity have been demonstrated to have a strong correlation with affective states, providing a physiologically based and objective method of evaluating mental health. However, issues with subject variability, hardware limitations, signal noise, and user comfort currently prevent EEG-based systems from being widely used in the real world.

Multimodal fusion overcomes the limitations of individual modalities by fusing external behavioural signals, such as speech and facial expressions, with internal physiological reactions captured by EEG and other biosignals. The fusion of complementary information sources in multimodal deep learning models enhances their resilience, generalization, and robustness, allowing them to effectively handle missing or noisy data [50, 64]. Fusion strategies enable models to learn cross-modal correlations, allowing one modality to compensate for the unavailability or unreliability of the other modality. In mental health settings, where emotional expression may be nuanced, erratic, or deliberately obscured, the significance of multimodal systems is amplified.

The studies examined indicate that multimodal systems yield results that are more comprehensible and beneficial in a clinical context compared to unimodal systems. This is because putting together different types of data makes the results easier to understand and more useful in a clinical setting. Multimodal frameworks improve the ability to recognize emotions and find depression by comparing computer predictions with physiological signs and observable behaviors. These types of systems are better for long-term monitoring because they can see how emotional patterns change over time instead of all at once.

There are still a lot of important research gaps, even with these changes. One significant issue highlighted in prior research is the scarcity of extensive, diverse, and meticulously annotated multimodal datasets, particularly those

that replicate authentic clinical scenarios. In lab settings, data collection is closely monitored, which makes it hard for trained models to work in the real world. This limits how well they can work outside of controlled environments. Differences in annotation standards and data collection protocols also make it hard to do cross-dataset evaluation and benchmarking.

Another big gap is that there isn't an adaptive and real-time multimodal emotion recognition system in use. Even though many existing studies have shown that they work well offline, there aren't many studies that focus on real-time inference, computational efficiency, and user interaction constraints. Furthermore, improving the incorporation of emotion recognition models into interactive conversational systems is crucial to optimize their effectiveness for ongoing mental health support and intervention.

These gaps show how important it is to have practical multimodal emotion-aware chatbot systems that can work well in the real world. To enable empathetic, context-aware, and ethically responsible interactions, these systems should integrate advanced emotion recognition with conversational intelligence. Incorporating multimodal emotion analysis into conversational agents could facilitate early mental health evaluation, ongoing emotional surveillance, and tailored user interaction, especially in contexts with restricted access to mental health practitioners [35, 40].

While multimodal deep learning has made significant progress in detecting depression and recognising emotions, the discussion as a whole shows that more work is needed to close the gap between experimental systems and deployable mental health applications. Addressing challenges related to dataset limitations, real-time implementation, and conversational integration is essential to transform these technologies into practical and easily accessible mental health support solutions.

6. Conclusion

The literature review scrutinized significant advancements in multimodal deep learning frameworks for mental health analysis, specifically emphasizing the automatic identification and early diagnosis of disorders such as depression, anxiety, and other affective conditions via the integration of various data types. The studies reviewed demonstrate that multimodal architectures, which integrate text, speech/audio, EEG, and visual indicators, effectively capture intricate behavioral and physiological signatures associated with mental health conditions.

6.1. Summary of Key Findings

This research presented a comprehensive virtual mental health support framework employing multimodal deep learning, which includes facial emotion recognition, speech and

text emotion analysis, EEG-based emotion and depression detection, and clinically validated psychological assessments. Research such as [4, 35] indicates that multimodal emotion recognition systems outperform unimodal systems by detecting diverse emotional cues across various modalities. CNN-based facial analysis is particularly adept at capturing expressive visual signals, as evidenced by research such as [38], while MFCC-driven recurrent models excel in learning temporal vocal patterns, as demonstrated by studies like [28]. EEG-based methods provide objective physiological indicators of emotional and depressive states [24, 27], whereas survey instruments such as PHQ-9 and GAD-7 offer reliable clinical validation [26, 49]. These findings collectively establish a robust foundation for adaptive and personalized mental health assessment.

6.2. Importance of Early and Accurate Diagnosis

Timely and accurate identification of emotional distress and depressive symptoms is crucial for improving mental health outcomes, as delayed diagnosis is associated with increased symptom severity and reduced intervention effectiveness. Prior research emphasizes that data-driven and multimodal approaches enable more reliable and continuous assessment by mitigating the subjectivity and limitations of self-report-based diagnosis [46, 41]. EEG signals serve as objective markers for affective disorders, as demonstrated in studies like [63], while behavioral cues from facial expressions and speech enhance the sensitivity to subtle emotional changes, as shown in research such as [31]. Because of this, multimodal frameworks are very important for making sure that interventions happen on time, that clinical decisions are made with all the facts in mind, and that mental health is managed better in the long term.

6.3. Call to Action for Future Research and Implementation

Recent advancements highlight the potential of multimodal deep learning for mental health support; however, further research is necessary for effective implementation. Subsequent research ought to concentrate on utilizing comprehensive and demographically diverse datasets, exploring advanced fusion methodologies, and developing customized models capable of adapting to factors such as age, cultural context, and distinct behavioral patterns, as evidenced in studies such as [35, 4]. Longitudinal clinical validation and integration with healthcare systems are imperative to guarantee reliability and usability [27]. Moreover, ethical factors like data privacy, openness, and user trust should be given the most importance and be at the heart of system design in mental health frameworks. For these frameworks to be turned into useful and responsible solutions for mental health, artificial intelligence researchers, clinicians, and mental health professionals need to work together over a long period of time.

Ethical Approval

Not applicable. This study did not involve any direct interaction with human participants and was conducted using publicly available datasets.

Consent to Participate

Not applicable. No primary data was collected from human participants, and all experiments were conducted using publicly available datasets.

Consent to Publish

All authors have read and approved the final manuscript and consent to its publication.

Data Availability

The datasets used in this study are publicly available, including DEAP [24], DAIC-WOZ [15], IEMOCAP [6], AffectNet [38], RAF-DB [34], FER2013 [14], SEED [63], AMIGOS [37], CMU-MOSI [60], and CMU-MOSEI [61].

Author Contributions

Aaditya Padte, Aniket Jadhav, Bhumi Batawale and Vedant Kale contributed to the conceptualization and survey of the proposed system and will also be responsible drafting the paper, and data processing. Sandip Shingade and Valmik Nikam provided project supervision, technical guidance, critical reviews and updating of the manuscript. All authors contributed to manuscript preparation and approved the final version.

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Competing Interests

The authors declare that they have no known competing financial interests or personal relationships that could have influenced the work reported in this paper.

Declaration of generative AI and AI-assisted technologies in the manuscript preparation process:

During the preparation of this work the author(s) used Quillbot in order to drafting the paraphrasing-tool. After using this paraphrasing-tool/service, the author(s) reviewed and edited the content as needed and take(s) full responsibility for the content of the published article.

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